

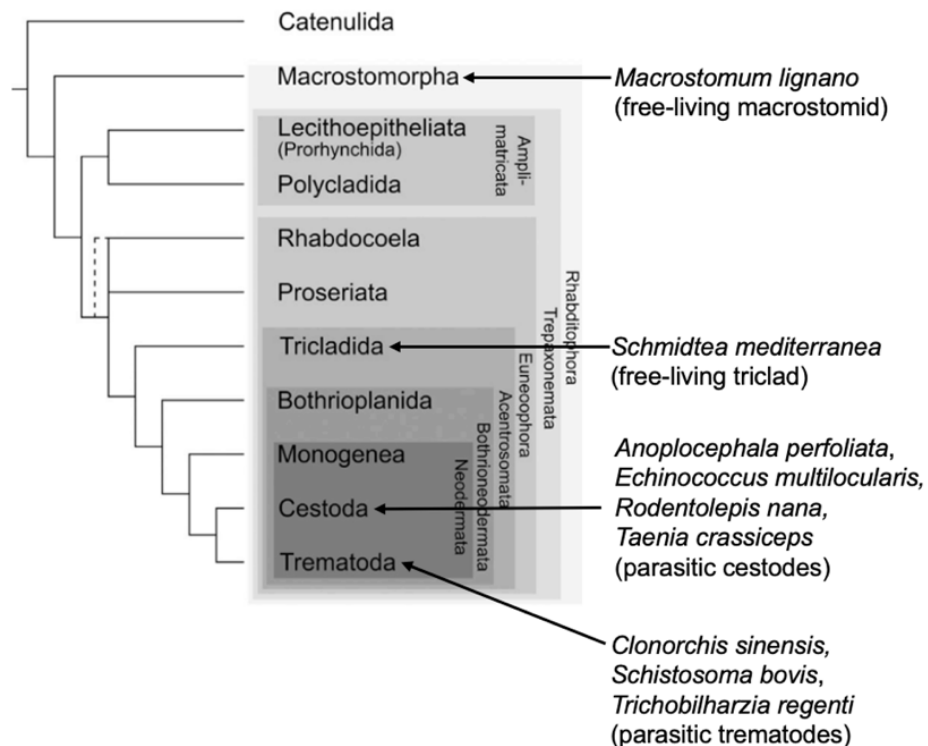
CANDIDATE CORRECTIONS TO MANUSCRIPT

- [Page 4]:
 - Figure 1: Phylogeny of metazoan species from which **HIF pathway gene** anchor sequences etc.

Protein sequences from 25 species across 13 animal phyla were used to construct **p-HMMs for HIF- α , EGLN and VHL ortholog mining** across the proteomes of 9 platyhelminth species.

Priapulida clades with Ecdysozoa, not Lophotrochozoa.

- [Page 5]:
 - Figure 2: **AMENDED TO**



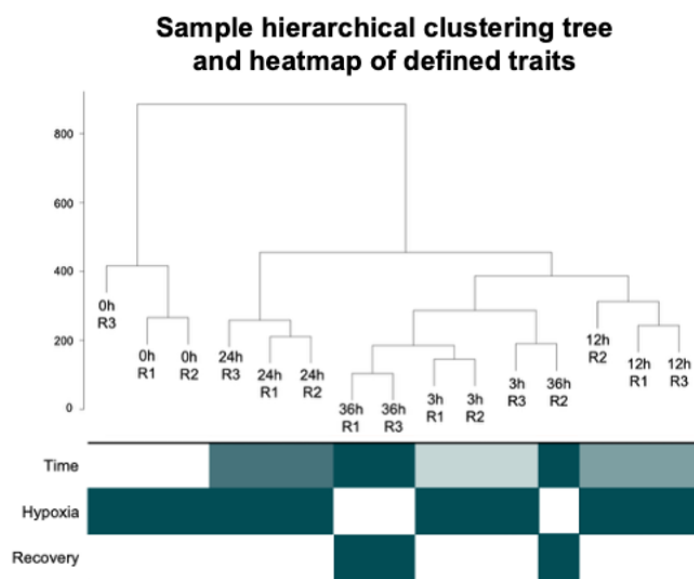
- [Page 10]: **FINAL PARAGRAPH AMENDED TO**

ATAC-seq consensus peakset construction

ATAC-seq consensus peakset construction was performed using DiffBind (Stark & Brown, 2011). A consensus peakset was subsequently generated using dba.count with minOverlap = 2, retaining peaks present in at least two libraries. Peak summits were re-centred using a 250 bp summit extension.

- [Page 14]:
 - **For HIF- α ortholog mining**, alignment using MAFFT iterative local-pair refinement was selected to maximise positional homology etc.
- [Page 16]:
 - A single candidate VHL ortholog in the **trematode** *Clonorchis sinensis*.
 - Subsequent maximum-likelihood phylogenetic inference placed the *Clonorchis* sequence **in a phylogenetic tree of high-confidence metazoan VHL sequences** as an isolated branch outside other metazoan VHL clades etc.

- Characteristic residue analysis showed a conserved central domain (positions 36-60) shared between the *Clonorchis* sequence and bivalve VHL-like proteins despite high divergence in flanking regions, supporting distant homology localised to a constrained functional motif **within the VHL β -domain**.
- [Page 18]:
 - Given that no orthologous sequence to the *Clonorchis* candidate VHL sequence could be identified by our HMMer search or forward BLASTp searches within the proteomes of the more basal-branching **cestodes**, *Schmidtea mediterranea* and *Macrostomum lignano*, these results suggest independent losses of VHL from **Cestoda**, Macrostomorpha and Tricladida at minimum within Platyhelminthes.
- [Page 21]:
 - Figure 11A: **AMENDED**



- Figure 11C: Heatmaps **beside** the module assignments etc.
- Figure 11D: Temporal trajectories of module expression showing mean FPKM **value** of genes etc.
- [Page 31]: These indirect sources of evidence are critical in a non-model metazoan with no transgenic technologies available, precluding the use of **CRE-reporter** constructs for directly testing the function and mapping the target genes of CREs.
- [Page 32]: Figure 18: Base-pair resolution view of hypoxia-specific Klf10/11 footprint in **h1SMcG0021003**.

A higher footprint ratio indicates stronger local depletion of Tn5 integration, consistent with increased transcription factor occupancy across cells. **The approximate footprint ratio presented here does not account for intrinsic Tn5 insertion bias, which was corrected by TOBIAS ATACorrect prior to ScoreBigwig and BINDetect analyses. These biases include preferential insertion at 9bp motifs enriched for cytosine at both edges (Zhang *et al.*, *NAR Genom Bioinform*, 2021). This accounts for why post-correction of these sequence-dependent biases, TOBIAS assigned occupancy scores of 0.454 at 0h and 0.622 at 3h hypoxia to this Klf10/11 motif. The lowest track shows nucleotide-resolution alignment of the predicted Klf10/11 binding-motif identified by TOBIAS BINDetect.**